

Medical Parasitology

Protozoa causing CNS diseases (I)

Common protozoa causing CNS diseases	▪ <i>Toxoplasma gondii</i> ▪ <i>Naegleria fowleri</i> ▪ <i>Acanthamoeba spp.</i>, ▪ <i>Entamoeba histolytica</i> (brain amoebic abscess) ▪ African <i>Trypanosoma</i> ▪ American <i>Trypanosoma</i> ▪ <i>Plasmodium falciparum</i>
---	--

Potentially Pathogenic free living amoebae

- These are small& free-living.
- Widely distributed throughout the world: in moist soil, fresh or stagnant water, air-conditioning units& decaying vegetation.
- They are considered as facultative parasites accidentally infecting man& causing serious infections.
- Two genera are of medical importance: Genus *Naegleria* and Genus *Acanthamoeba*.

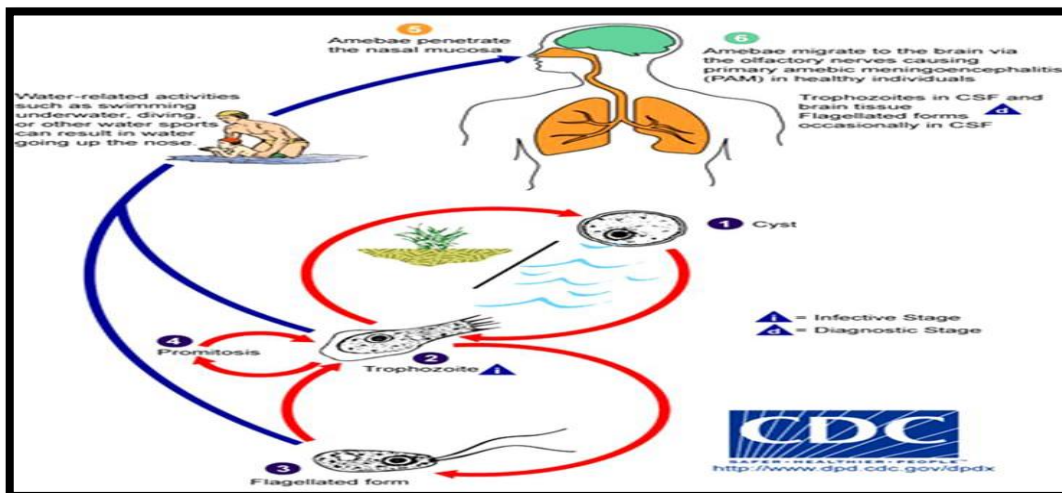
Naegleria fowleri

Disease	Naegleriasis or Primary amoebic meningo-encephalitis (PAM)
Habitat	Found in warm freshwater (lakes, rivers, ponds) and in moist soil. On infection of man, it is present in the central nervous system.
Infective stage& Diagnostic stage	Amoeboid form trophozoite (No autoinfection)
Mode of infection	Infection occurs through <u>the nasal route</u> by: 1. Swimming or diving in contaminated water; this is the most common way of infection.

	2. Inhalation or aspiration of contaminated air.
	3. Sniffing contaminated water.

Life cycle:

- *Naegleria fowleri* is found in nature in warm water bodies as ameboid and ameboflagellate trophozoites. Cysts also occur in nature, but not in human infections.
- Man is infected during swimming or bathing in contaminated water.
- The amoeboid form (Infective stage) trophozoites are neurotrophic, enter the nose, invades the nasal mucosa and cribriform plate, travel via the olfactory nerve reaching the brain. These forms are active, multiply by binary fission, ingest red and white blood cells and cause tissue destruction. Cysts usually are not seen because the infection is so rapid and fatal that the patient dies before trophozoites encyst to the cysts.



Pathogenesis and clinical picture:

- Amoeboid trophozoites reach the brain (the nose, the nasal mucosa, the cribriform plate, the olfactory nerves, brain and meninges) causing acute Hemorrhagic necrotizing Meningoencephalitis.
- The course of PAM is dramatic.
- Incubation period: 2-4 days after exposure and swimming in freshwater.
- There is fever, rhinitis, pharyngitis, headache followed by nausea, projectile vomiting, stiffness of the neck (meningitis) and convulsions.
- Disturbance in smell or taste may occur.
- The patient enters in coma and death within 3-6 days.
- Prognosis: Is very poor, most cases are fatal and death occurs in more than 95% of cases

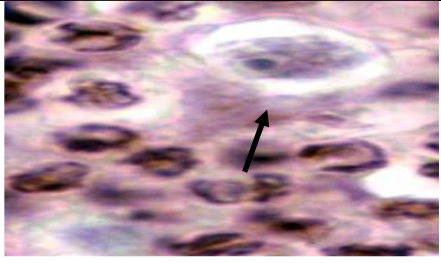
Diagnosis:

I. Clinical diagnosis: Recent history of swimming in thermal, stagnant water or contact with fresh water, mud or dust, 2- 6 days prior to the onset of symptoms and clinical manifestations. The patient is usually child or young adult.

II. Laboratory diagnosis:

Cerebrospinal fluid (CSF) examination: Obtained by lumbar puncture. To differentiate PAM from bacterial or viral encephalitis.

- Macroscopically: It appears cloudy, purulent or bloody.
- Microscopic examination: Shows amoeboid form, leucocytosis (neutrophils) & RBCs, but no bacteria.
- Chemically: Increased protein and decrease glucose levels.
- Culture: On non-nutrient agar plate seeded with *Escherichia coli*.
- Suspension (Enflagellation test): in fresh water; shows transformation of amoeboid forms into flagellate forms rapidly within 2 hours.

Amoeboid trophozoite Shape elongated, 10-15 μm in diameter, has a clear ectoplasm and a granular endoplasm with single large nucleus which has a large central karyosome but no peripheral chromatin.	
--	---

III. Postmortem examination of brain tissues: Reveals extensive hemorrhage with amoeboid trophozoites but not cysts.

Treatment:

Amphotericin B (the drug of choice): 1 mg/Kg/day for 9 days.

It is given intravenous & intrathecal until CSF is cleared

Prevention and control:

1. Avoid swimming in contaminated water
2. Proper chlorination of public water supplies and water of swimming pool.

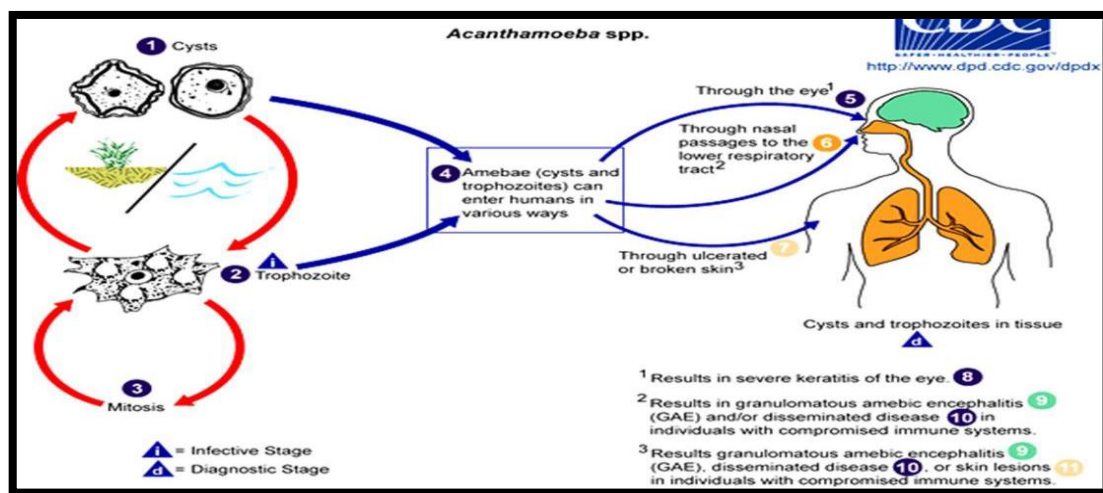
Acanthamoeba Spp.

Disease	1. Granulomatous amoebic encephalitis (GAE). It occurs mostly in immunocompromised persons 2. <i>Acanthamoeba</i> keratitis (AK) in healthy persons.
----------------	---

Habitat	CNS, skin, eye, lung or genitourinary tract
Infective stage	Trophozoite, Cyst
Diagnostic stage	Trophozoite
Mode of infection	1. CNS infection is blood borne; it occurs secondary to infection in skin or lower respiratory tract. 2. Blood carries the parasite from the primary lesions to the CNS & may be to other organs as kidney, uterus& pancreas.

Life cycle:

1. The life cycle is simple and consists of perpetuation between the active trophozoite and the resting resistant cyst stages.
2. Under unfavorable conditions (lack of food, extremes in osmolarity, pH and temperatures), amoebae switch into resistant cyst stage.
3. *Acanthamoeba* spp. does not have an ameboflagellate form, and cysts can be found in human infections. Infections occur more frequently in debilitated or chronically ill individuals.
4. Both cysts and trophozoites are the infective stages.



Pathogenesis and clinical picture:

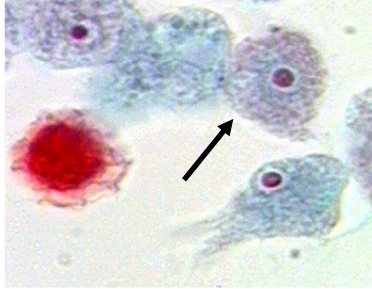
- Chronic infection characterized by focal granulomatous lesions in the brain and in other affected organs. Both trophozoites and cysts are found in these lesions
- The incubation period is not known (few weeks to months or even years).

- There is change in mental state, headache, convulsions, nausea, vomiting & neck rigidity
- In AIDS patients, the disease may be severe & similar to infection with *Naegleria*.

Diagnosis:

I. Clinical diagnosis: Granulomatous amoebic encephalitis (GAE) is rarely diagnosed before death. Most cases have been diagnosed post-mortem or shortly before death.

II. Laboratory diagnosis: CSF examination demonstrates trophozoites.

<i>Acanthamoeba trophozoites</i> Slightly larger than that of <i>Naegleria</i> spp., variable in shape and size (10-40 μm). has slender, spine like pseudopodia (Acanthopodia). The cytoplasm is finely granular containing single nucleus with a large dense central karyosome.	
--	--

Treatment:

1. Excision of focal lesion + Ketoconazole
2. Penicillin & chloramphenicol.

Prevention and control:

The preventive measures to control GAE are difficult as the amoebae are present in air, soil and water.

<i>Naegleria</i> Meningoencephalitis	<i>Acanthamoeba</i> Encephalitis
Mainly in Children & young adults	Mainly in Debilitated, chronically ill with low immunity
<u>Lesion:</u> Diffuse meningoencephalitis	<u>Lesion:</u> Focal, granulomatous, space-occupying lesion.
<u>Runs</u> rapidly fatal course (death within 3-6 days)	<u>Runs</u> sub-acute or chronic course (lasts for weeks, months or years)
<u>History</u> of swimming in natural water or swimming pools	<u>Not</u> strongly associated with swimming
Infection occurs through:	Infection occurs in:

The nasal route → cribriform plate → olfactory nerve → brain.	Lower respiratory tract, ulcerated skin or mucosa → blood stream → CNS
--	--

Amoebic brain abscesses

- Although amoebic brain abscess is a rare form of invasive amoebiasis (0.6% to 8.1%), when present, it is frequently lethal.
- This disorder always begins with the infection of the colon by *Entamoeba histolytica* trophozoites, which then travel to extra-intestinal tissues through the blood stream.
- When trophozoites invade CNS, amoebic brain abscesses are produced.
- Incubation period starting from a few days to several months.
- It is characterized by an abrupt onset of symptoms & rapid progression and early death if untreated.
- Amoebic brain abscess is mainly single, small and is located in the cerebral hemisphere.
- The symptoms and signs simulate a brain tumor
- Diagnosis: Clinically difficult as it simulates a brain tumor.
- Serological tests: Indirect Hemagglutination test and ELISA
- Radiological investigation: CT and MRI can be used to diagnose the presence or absence of a brain abscess with a certainty of 100%.
- Accurate diagnosis of amoebic brain abscess is often done by identification of *Entamoeba histolytica* trophozoite in a biopsy, by immunohistochemistry and immunofluorescence with specific antibodies.
- Treatment: A combination of emetine and metronidazole as both drugs enter nervous tissue
- Prevention and control:
 1. Health education.
 2. Safe water supply.
 3. Proper sewage disposal.
 4. Treatment of cases.
 5. Repeated examinations of food handlers.
 6. Insect control.

Protozoa causing CNS diseases (II)

Objective ILOs

- List the most common protozoa causing CNS diseases (African *Trypanosoma*, American *Trypanosoma*, *Plasmodium falciparum*).
- Recall the life cycle, the sources of infection, the infective and diagnostic stages of the selected parasite (African *Trypanosoma*, *Plasmodium falciparum*).
- Describe disease pathogenesis and main clinical presentations of the selected parasite (African *Trypanosoma*, *Plasmodium falciparum*).
- Describe laboratory diagnosis, imaging and pathological studies of lesions caused by the selected parasite (African *Trypanosoma*, *Plasmodium falciparum*).
- Recall treatment and prevention of the selected parasite (African *Trypanosoma*, *Plasmodium falciparum*).

African trypanosomes

(*Trypanosoma brucei gambiense*, *Trypanosoma brucei rhodesiense*)

Disease	African trypanosomiasis (sleeping sickness) <u><i>Trypanosoma gambiense</i></u> : West African trypanosomiasis <u><i>Trypanosoma rhodesiense</i></u> : East African trypanosomiasis
Geographical distribution	Tropical Africa
Habitat	▪ <u>During the early stages of the disease</u>; Trypanosomes are found extracellular in the peripheral blood, lymph, tissue spaces of various organs of R.E.S. (Liver, Lymph nodes, Bone marrow, Spleen). ▪ <u>In the terminal stages</u>; Trypanosomes are found in Central Nervous System.
Definitive host (D.H.)	Man
Intermediate host (I.H.) Vector	<i>Glossina spp.</i> (Tse tse fly) <u><i>Trypanosoma gambiense</i></u> : <i>Glossina palpalis</i> <u><i>Trypanosoma rhodesiense</i></u> : <i>Glossina morsitans</i>

Infective stage (I.S.)	Metacyclic trypomastigote (short stumpy trypomastigote with undulating membrane but without free flagellum).
Diagnostic stage (D.S.)	Polymorphic trypomastigotes
Mode of infection	1. Bite of infected <i>Glossina spp.</i> inoculating the metacyclic trypomastigotes with saliva while taking their blood meals (anterior station). 2. Rarely by blood transfusion or congenital transmission.
Life cycle	

Pathogenesis and Clinical picture:

1. Trypanosoma chancre (skin):

Large, red, painful, indurated and rubbery swelling due to multiplication of the parasite (Trypomastigotes) at the site of the bite

2. Hemolymphatic stage (lymph node):

Trypanosomes are present in the blood, lymph nodes and bone marrow, then multiply and cause:

-Toxic manifestations: Irregular fever, headache, muscle wasting, joint pain and myxedema.

-Lymphocytic hyperplasia: Leads to hepatosplenomegaly and generalized lymphadenopathy.

Winterbottom sign: Enlargement of superficial lymph node in posterior triangle of neck.

-Bone marrow affection (pancytopenia): Leads to anemia, leucopenia and thrombocytopenia

3. Meningo-encephalitic stage (sleeping sickness stage):

- It occurs at the end of first year of infection.
- Trypomastigotes invade CNS, leads to perivascular lymphocytic infiltration of cerebral vessels, brain damage, meningitis and hemorrhagic encephalopathy.
- Clinically manifested by: Aggravation of symptoms of hemolymphatic stage, Insomnia, mood change (dullness, apathy), motor and sensory disorders (hyperesthesia, slurred speech, abnormal gait), convulsions, Epilepsy, disturbed mentality, paralysis. Lastly, the sleeping stage develops and the patient falls into coma and finally death due to disease or intercurrent infection.

Clinical features of Gambian and Rhodesian disease are similar, but they vary in severity and duration:	
Gambian Sleeping sickness (<i>T. b. gambiense</i>)	Rhodesian Sleeping Sickness (<i>T. b. rhodesiense</i>)
▪ <u>Parasite</u>: Less virulent ▪ <u>Disease progresses</u>: Slowly (Chronic) ▪ <u>Parasite in blood</u>: Scanty (Low parasitaemia). ▪ Typical sleeping sickness symptoms. ▪ <u>Animal inoculation</u>: Refractory	▪ <u>Parasite</u>: Extreme virulence ▪ <u>Disease progresses</u>: Rapidly (Acute) ▪ <u>Parasite in blood</u>: Plenty (High parasitaemia). ▪ Usually fatal before sleeping sickness symptoms appear. ▪ <u>Animal inoculation</u>: Susceptible with posterior-nuclear shift.

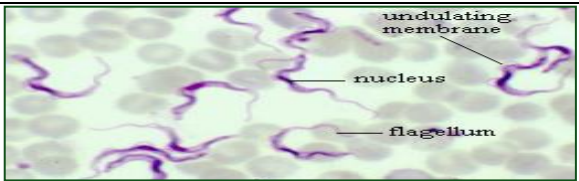
Diagnosis:

I. Clinical diagnosis: History (residence or traveling to endemic area) and Clinical picture.

II. Laboratory Diagnosis:

A. Direct methods: To demonstrate the parasite, early (from chancre aspirate, blood, lymph node, bone marrow), late (in CSF) BY

a. Microscopic examination of fresh unstained or Giemsa stained films: polymorphic trypomastigote. **It is most sensitive in early stages of disease, when the number of circulating parasites is highest.**

Polymorphic trypomastigote	
Long and slender (30μ)	
Short and stumpy (18μ)	
Intermediate forms (22μ)	

- b. Culture: Blood on NNN medium (Epimastigotes form).
- c. Animal inoculation: Intraperitoneal injection of mice or hamsters followed by blood examination to see polymorphic trypomastigotes. Done for *Trypanosoma rhodesiense* to demonstrate forms with posterior-nuclear shift.
- d. CSF examination: It is a must, especially in late stage. It shows trypomastigotes, increase cell count, increase IgM, lymphocytes and Morula cells of Mott which is a vacuolated plasma cell.

B. Indirect methods:

- a. Detecting anti-*Trypanosoma* Abs by serological methods (ELISA, IHA). But, can't distinguish between current and previous infections.
- b. Card Agglutination Trypanosomiasis Test (CATT): It is a simple & rapid test for detection of circulating antigens in the blood of the patient. It is useful in surveys specially.
- c. Molecular techniques (e.g. PCR).

Treatment:

- 1. Early stage (Haemolymphatic Stage):** Pentamidine, Suramin (Antrypol).
- 2. Late stage (CNS invasion):** Melarsoprol (Mel B), and Eflornithine (DMFO). Both can cross blood –brain barrier but have marked side effects. Nifurtimox (Lampit) is used in combination with (DMFO) in melarsoprol resistance.

Prevention and control:

- 1. Treatment of patient.
- 2. Control of *Glossina* (vector).
- 3. Protection against vector as by skin repellents.
- 4. Chemoprophylaxis in endemic areas by Pentamidine at 4-6 months intervals.

Plasmodium falciparum (malignant malaria)

Disease	Malignant malaria
Habitat	Blood

Definitive host (D.H.)	Female <i>Anopheles</i> mosquito
Intermediate host (I.H.)	Man
Mode of infection and Infective stage	1. Bite of infected female <i>Anopheles</i> mosquitoes (sporozoite) 2. Blood transfusion from a donor with erythrocytic stages (merozoite) 3. Rarely due to the common use of syringes (merozoite) 4. Congenital transmission (merozoite)
Diagnostic stage (s)	Only rings and gametocytes are seen in peripheral blood. Early and mature Trophozoites, and Schizonts are retained in capillaries of internal organs
Life cycle	

Pathogenesis and clinical picture of *Falciparum* Malaria

- The characteristic fever peaks of malaria result from simultaneous rupture of many erythrocytes, with the release of parasites debris and malarial pigments causes a rise in temperature (paroxysmal attacks).
- Hepatomegaly, splenomegaly and bone marrow hyperplasia: Malarial pigments are engulfed by macrophages in liver, spleen, bone marrow. This results in hypertrophy of organs.
- Hemolytic anemia: destruction of erythrocytes. Anemia is severe in *P. falciparum* due to:
 - Heavy parasitaemia (large no. of merozoites are released from each RBC).
 - Infects RBCs of all ages.

- Hemolysis of parasitized and non-parasitized red cells due to immunosensitivity.
- Late parasite stages, called trophozoites and schizonts (mature forms have the ability to adhere to capillary and endothelium, a process that is called cytoadherence and adhere to non-infected RBCs (called rosetting). The resulting sequestration of infected erythrocytes (IRBC) leads to alterations in microcirculatory blood flow, metabolic dysfunction, and, as a consequence, many of the manifestations of severe falciparum malaria
- In Brain: This process leads to cerebral anoxia, development of malaria granulomas and punctate hemorrhages leading to malarial encephalitis and meningoencephalitis
- Incubation period in most cases varies from 7 to 10 days
- The classical malarial paroxysm: Cold stage lasts 15-60 minutes and is characterized by a feeling of cold and shivering. This is followed by the hot stage, with fever from 39-41.5°C, lasting 2-6 hours, also associated with flushed, dry skin, and often headache, nausea, and vomiting. At the end of it, the fever drops rapidly and the patient sweats profusely over a period of 2-4 hours. **falciparum may not exhibit classic paroxysms (continuous fever)**

- Complications:

- Cerebral malaria
- Dysenteric malaria
- Hypoglycemia
- Acidosis
- Pulmonary edema
- Black water fever
- Algid malaria

***Cerebral malaria**

- Cerebral malaria is defused encephalopathy where focal symptoms are unusual
- Histopathology include petechial hemorrhages in the brain parenchyma, ring hemorrhages in white matter and Durck's granuloma's
- Clinical picture: The disease starts suddenly or slowly, but rapidly progress to cause impaired consciousness, high fever, chill, generalized convulsions, shock, coma that persists for 24 -72 hours and death within days or weeks.
- This encephalopathy is potentially reversible

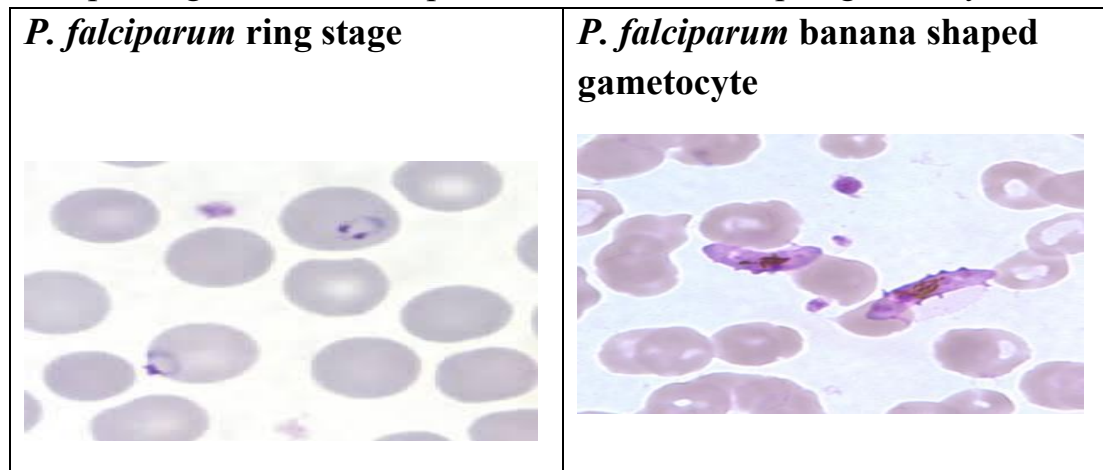
***Diagnosis:**

I. Clinical diagnosis: In severe malaria (*Plasmodium falciparum*), clinical findings (confusion, coma, neurologic focal signs, severe anemia, respiratory difficulties) increase the index of suspicion for malaria.

II. Laboratory diagnosis:

1. Direct method:

Malaria Blood Smear the gold standard for laboratory confirmation of malaria. A blood specimen collected from the patient is spread as a thick or thin blood films, stained with a Giemsa and examined with a 100X oil immersion objective. Erythrocytes containing multiple ring forms and the presence of banana-shaped gametocytes



2. Indirect methods:

Serology: ELISA and IFA test, immunofluorescence (IFA) or enzyme-linked immunosorbent assay (ELISA). Serology does not detect current infection but rather measures past exposure.

Rapid diagnostic tests most often use a dipstick or cassette format and provide results in 2-15 minutes.

3. Molecular technique: Polymerase Chain Reaction (PCR)

Treatment of cerebral malaria

1. Admit ICU for monitoring, surveillance and support treatment.
2. The intravenous administration of artemisinin derivatives, artesunate and quinine are the drugs of choice

Prevention and control:

1. Control of mosquitos
2. Personal protection against mosquitos' bites as bednets, repellent, protective clothes, and chemoprophylaxis if travelling to endemic area.
3. Treatment of patients.